DATA CARD

Reliable Chest X-Ray AI | NIH ChestX-ray14

ACCESS DATE 27 October 2026
MODALITY Chest X-ray images + metadata

112,120	30,805	53.84%	46.16%	15
Image records	Unique patients	Normal (60,361)	Abnormal (51,759)	Original findings

1 SOURCE, CONTENT & ACCESS

Provider: **NIH Clinical Center**. Official source: NIH ChestX-ray dataset page. The project uses **112,120 PNG images from 30,805 unique patients**. Metadata fields: image name, finding labels, follow-up index, patient ID, age, sex, view position (AP/PA), image width/height, and pixel spacing (x/y). Recorded image dimensions vary: width 1,143–3,827 px and height 966–4,715 px.

2 LABEL GENERATION & BINARY DEFINITION

The 15 original findings were automatically extracted from associated radiology reports using an NLP labeler. These are **weak labels**, not independently confirmed ground truth for every image. **Normal = "No Finding" (0):** 60,361 images (53.84%). **Abnormal = one or more other findings (1):** 51,759 images (46.16%). An image with multiple abnormal findings counts as **one Abnormal example** in this binary task; this simplifies the original multi-label problem.

3 VIEW POSITION & ACQUISITION

AP: 44,810 images (40.0%); beam travels anterior-to-posterior and is often used for portable/bedside imaging. **PA:** 67,310 images (60.0%); beam travels posterior-to-anterior and is commonly acquired upright. These acquisition conditions are not interchangeable. Evaluate model performance by view position; descriptive label differences alone do not prove acquisition bias or model error.

4 PATIENT & DEMOGRAPHIC STRUCTURE

Repeated examinations: 13,302 patients (43.18%) have more than one image; one patient has up to 184 images. Split by **patient_id**, not randomly by image, to reduce patient-level leakage. **Age:** Normal mean 45.48 years (median 47); Abnormal mean 47.96 (median 50). Fourteen rows record age 0 and need an explicit handling decision. Sex proportions are broadly similar in the reviewed binary groups, but subgroup representation and performance should still be assessed.

5 KNOWN LIMITATIONS & DATA QUALITY

Automatically extracted labels may be missing, incorrect, or imperfectly aligned with the visible image. The binary target collapses multiple abnormal disease categories into one class and can hide clinically meaningful distinctions. Class proportions are moderately imbalanced (53.84% Normal vs 46.16% Abnormal). Repeated images per patient and AP/PA acquisition differences can affect evaluation. The audited metadata CSV had no missing values or duplicate metadata rows; this does not guarantee error-free images, labels, or clinical records.

6 USAGE CONDITIONS, ETHICS & INTENDED USE

Follow NIH usage guidance: acknowledge the NIH Clinical Center, link the official source, and cite the original publication. NIH describes use as unrestricted subject to attribution; the referenced source does not specify a separate standardized Creative Commons license. The public release is de-identified: do not attempt re-identification. Assess limitations and potential bias responsibly. **Intended use:** academic research and model evaluation only. This is a **research prototype, not a clinically validated diagnostic tool**; it must not replace radiologists or guide care without appropriate clinical validation and oversight.

ORIGINAL FINDING LABELS — MULTI-LABEL DISTRIBUTION

ORIGINAL FINDING LABEL	COUNT	%
No Finding	60,361	53.84%
Infiltration	19,894	17.74%
Effusion	13,317	11.88%
Atelectasis	11,559	10.31%
Nodule	6,331	5.65%
Mass	5,782	5.16%
Pneumothorax	5,302	4.73%
Consolidation	4,667	4.16%
Pleural Thickening	3,385	3.02%
Cardiomegaly	2,776	2.48%
Emphysema	2,516	2.24%
Edema	2,303	2.05%
Fibrosis	1,686	1.50%
Pneumonia	1,431	1.28%
Hernia	227	0.20%

Percentages are per image and sum to more than 100% because one image may have multiple findings. These original labels are distinct from the binary target.

ORIGINAL PUBLICATION

Wang, X., Peng, Y., Lu, L., Lu, Z., Bagheri, M., & Summers, R. (2017). *ChestX-ray8: Hospital-scale Chest X-ray Database and Benchmarks on Weakly-Supervised Classification and Localization of Common Thorax Diseases.* IEEE CVPR. doi:10.1109/CVPR.2017.369.

ADDITIONAL DATA AUDIT

Follow-up index: mean 8.57, median 3, range 0–183; it is an index, not elapsed time. The reviewed metadata had no missing values or duplicate metadata rows.

OFFICIAL SOURCE

NIH ChestX-ray dataset page

DUPLICATE IMAGES AND DATA LEAKAGE RISK

Dataset update: 09 October 2026

The NIH ChestX-ray dataset update dated October 9, 2026, reported several pixel-identical image pairs with different filenames. Four reported pairs span the official training and testing splits, creating a potential data leakage risk. Consequently, the official split cannot be assumed to be entirely free of duplicate-image leakage. We will audit image duplicates and patient-level overlap before model development, document the duplicate-handling policy, and evaluate the final model on a clean, independent test set. The reported duplicates and their potential impact will be documented as dataset limitations.

FOUR REPORTED TRAIN–TEST PAIRS

Train `00027864_003` ↔ Test `00005681_024`
Train `00012388_000` ↔ Test `00017214_007`
Train `00027864_000` ↔ Test `00028328_000`
Train `00012388_001` ↔ Test `00017214_007`

Proposed handling (verify files first): exclude from Test `00005681_024`, `00017214_007`, and `00028328_000`. Review the within-Train duplicate pair `00012388_000` / `00012388_001` and retain only one copy.

Related data-quality controls: split by `patient_id`, not by image; treat report-extracted labels as weak labels; evaluate AP and PA subgroups separately; and define a policy for records with age = 0. Proposed duplicate removals remain unconfirmed until image files and metadata are checked.